SHORT ANSWER TEST

ASSESSMENT

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The smartphone camera captures images with low spatial resolution which means fewer pixels are used to represent the image. The problem is mainly caused by insufficient sampling, and more sampling preserves more details. Increasing the number of pixels improves image resolution and a high resolution camera is suitable and the camera should use higher spatial sampling rate.

2) Poor and low lighting can cause insufficient brightness and increase noise in the image. Histogram equalization can improve image brightness & contrast. A noise reduction & median filtering could be used for reducing unwanted & impulse noise. The enhancement makes important objects more visible and the procesed image becomes easier to analyze. So enhancement is suitable for night surveillance.

3) Computers require discrete digital values for processing medical images. Quantization converts continous intensities into finite levels and each

intensity value is assigned. It allows images to be stored digitally and makes image processing & analysis possible. Proper quantization reduces unnecessary intensity information.

4) Contrast enhancement can improve the uneven lighting by improving visibility. Histogram equalization can redistribute intensity values and increases contrast between defects & background. Adaptive histogram equalization can handle local lighting and the defects become more prominent after enhancement, which helps to detect imperfections.

5) Image segmentation divides images into meaningful regions. Roads, vehicles and pedestrians can be separated & identified. From the segmentation, the system obtains information about object location and helps vehicles ~~outlan~~ understand the surroundings. It supports decisions like braking and steering, therefore segmentation improves autonomous driving safety.

Cracks usually appear as sharp intensity changes and edge detection highlights these changes. Canny edge detection method detects strong boundaries & thin edges, therefore cracks become visible. It helps to locate the defective areas and it analyzed automatically. This automated detection reduces manual inspection effort.

7) A suitable filtering technique would reduce the noise like median filter which replaces each pixel with median of nearby pixels, ~~se~~ reducing impulse noise. The filter also preserves important edges as well & tumor boundaries remain clearer. The processed image becomes easier to examine & improves analysis.

8) Object detection identifies the vehicles ~~—~~ & modern methods like YOLO detect vehicles easily. Bonding ~~to~~ box is given to each vehicle and information is provided. Traffic density & congestion is monitored and accidents and unusual traffic can also be detected. Therefore, improves traffic management.

9) It is called motion blur & occurs when object or camera moves during exposure. Movement spreads image ~~for~~ across multiple pixels & reduces sharpness of the image. Motion deblurring can be used & which estimates & reverses the blur effect. Image restoration can also improve sharpness & shutter speed for a better & clearer image.

10) Image classification can automatically classify product images by learning visual patterns like shape, texture & colour features. An ML model can classify products as good or defective. This ~~reduced~~ reduces need for manual inspection & improves consistency in quality control. So, the image classification supports automated quality inspection.

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